

the selected microcontroller and leave resource margin for additional BMS tasks.

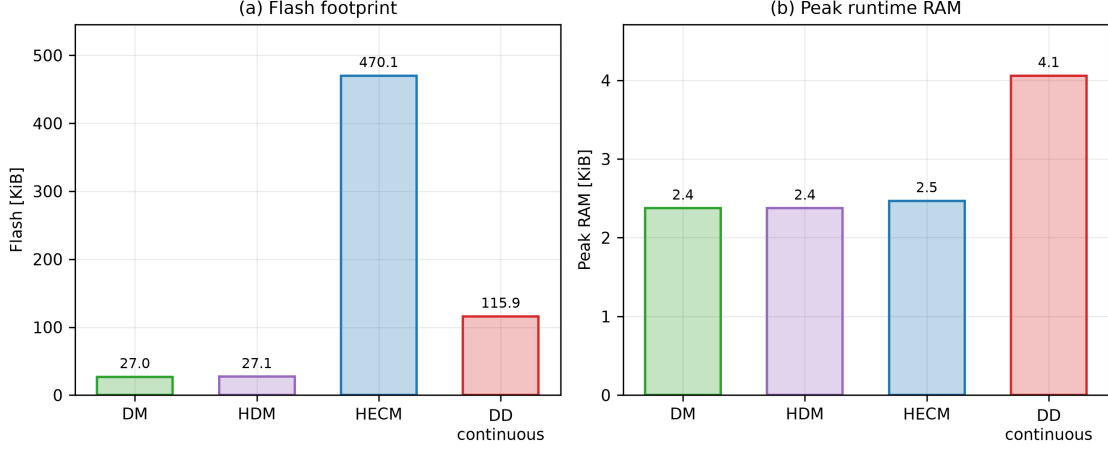


Figure 6.19: Embedded memory requirements of the isolated STM32 SOC implementations. Panel (a) reports compiled flash occupancy. Panel (b) reports peak runtime RAM, combining statically allocated variable storage, maximum dynamic allocation, and call-stack demand. DD denotes continuous-state execution.

The DD execution-mode comparison in Figure 6.20 makes the deployment trade-off explicit. The same pruned network weights are used in every mode. Exact rolling-window inference preserves the primary software semantics but requires approximately 724 ms median inference time and 67.3 KiB peak runtime RAM because the complete 2024-sample sequence is evaluated for every output. This occupies 72.4% of the one-second sampling period. Continuous-state execution evaluates only the new recurrent step and reduces latency to approximately 0.425 ms and peak RAM to 4.1 KiB while retaining similar nominal error over the six sequences. Periodically resetting the recurrent state has comparable memory and latency but creates much larger transient errors. The mean of the cellwise maximum dataset-SOC errors is approximately 29 percentage points, and individual cells reach approximately 39 percentage points. Each reset removes the temporal context accumulated before that point, whereas continuous forwarding preserves it.

Continuous-state forwarding therefore provides the practical DD hardware configuration. Its recurrent semantics differ from the rolling-window benchmark and must be validated as a distinct deployment mode. Appendix Figure A.2 provides the full latency distributions for all three modes.